Networks and Complexity

Solution 21-1

This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB

Ex 21.1: Laplacian construction [Lvl. 1]

Construct the matrices $\mathbf{A}$, $\mathbf{L}$, $\mathbf{L_{rw}}$, $\mathbf{L_{sym}}$ for a cycle of four nodes.

<u>*Solution*</u>

The adjacency matrix is

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}. \tag{1}$$

We compute the Laplacian as $\mathbf{L} = \mathbf{K} - \mathbf{A}$ where $\mathbf{K} = 2\mathbf{I}$ because every node has degree two. This yields

$$\mathbf{L} = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}. \tag{2}$$

To find the random-walk Laplacian we divide every row by the degree

$$\mathbf{L_{rw}} = \begin{pmatrix} 1 & -1/2 & 0 & -1/2 \\ -1/2 & 1 & -1/2 & 0 \\ 0 & -1/2 & 1 & -1/2 \\ -1/2 & 0 & -1/2 & 1 \end{pmatrix}. \tag{3}$$

To find the symmetric Laplacian, we compute $\mathbf{L_{sym}} = \mathbf{K}^{-1/2}\mathbf{L}\mathbf{K}^{-1/2}$.

We have

$$\mathbf{K}^{-1/2} = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 0 & 1/\sqrt{2} \end{pmatrix}, \tag{4}$$

and hence

$$\mathbf{L_{sym}} = \begin{pmatrix} 1 & -1/2 & 0 & -1/2 \\ -1/2 & 1 & -1/2 & 0 \\ 0 & -1/2 & 1 & -1/2 \\ -1/2 & 0 & -1/2 & 1 \end{pmatrix}. \tag{5}$$

Since the random walk Laplacian was already symmetric it is identical to the symmetric Laplacian. This happens in every network where the components are degree regular.